Percolation Experiment Lab Notebook[†]

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1 Objective

The objective of this experiment is to study the percolation transition by measuring the electrical resistance of a lattice network as bonds are progressively removed. The experiment aims to explore the relationship between resistance and the probability of broken bonds, and to analyze the nature of the phase transition.

2 Introduction

Percolation theory is used to model the behavior of connected clusters in a random medium. In this experiment, we consider bond percolation on a 2D square lattice, where electrical connectivity is gradually disrupted by removing bonds. The percolation threshold, p_c, marks the critical probability at which a spanning cluster ceases to exist, leading to a dramatic increase in resistance.

3 Bond percolation

Bond percolation examines connectivity in grid-like structures by treating edges (bonds) as probabilistic pathways. At a critical probability threshold p_c these systems undergo a phase transition from isolated clusters to spanning connectivity—a phenomenon with applications in material science, epidemiology, and network analysis.

3.1 Model Definition

- **Lattice structure**: Regular d-dimensional lattice with:
  - Vertex set V
  - Edge set E
- **Bond states**: Each edge $e \in E$ independently designated as:
  - Open (probability p)
  - Closed (probability $1 - p$) under Bernoulli measure μ_p

[†]Use Digital copy for the colored plots and diagrams.

- **Connectivity**: Open cluster C_v at vertex v contains all vertices reachable via open paths

3.2 Critical Phenomena

3.2.1 Percolation Probability

$$\theta(p) = \mu_p(|C_v| = \infty) \tag{1}$$

Probability that vertex v belongs to an infinite cluster.

3.2.2 Critical Threshold

$$p_c = \inf\{p \in [0,1] : \theta(p) > 0\} \tag{2}$$

Lattice Type	p_c (bond)
2D square	0.5
(1+1)-d directed square	0.644701
3D cubic (nearest neighbor)	0.287338

3.3 Phase Behavior

- **Subcritical** ($p < p_c$):
  - All clusters finite almost surely
  - Exponential decay: $\mu_p(0 \leftrightarrow x) \leq e^{-|x|/\xi}$ (correlation length ξ)
- **Critical** ($p = p_c$):
  - No infinite clusters
  - Power-law decay: $\mu_p(|C_v| \geq n) \sim n^{-1/\delta}$
  - 2D conformal invariance (related to SLE theory)
- **Supercritical** ($p > p_c$):
  - Unique infinite cluster emerges
  - Positive open edge density ensures long-range connectivity

3.4 Mathematical Formalism

- **Product measure**:

$$\mu_p = \prod_{e \in E} \text{Bernoulli}(p) \tag{3}$$

- **Dual lattices**: In 2D square lattices:

$$p_c + p_c^* = 1 \quad \text{(self-dual at } p = 0.5) \tag{4}$$

3.5 Critical Exponents & Universality

Universal scaling near p_c:

$$\xi \sim |p - p_c|^{-\nu} \tag{5}$$

$$\theta(p) \sim (p - p_c)^{\beta} \tag{6}$$

Dimension	ν	β
2D	4/3	5/36
$d \geq 6$ (mean-field)	1/2	1

4 Experimental Setup

4.1 Materials Required

- Graph paper with 9x9 and 11x11 lattice grids
- Pencil and eraser
- Multimeter for resistance measurement
- Conductive material (such as graphite or metallic connections)
- Ruler

4.2 Methodology

1. A square lattice of size 9×9 and 11×11 was drawn on graph paper.
2. Each intersection was enumerated systematically to track connectivity.
3. Bonds (lattice connections) were progressively erased at regular intervals to simulate bond percolation.
4. The resistance between two opposite ends of the lattice was measured after each stage of bond removal.
5. The probability of bond removal was calculated as:

$$p = \frac{\text{Bonds removed}}{\text{Total bonds}} \tag{7}$$

6. A graph of resistance (R) vs. probability (p) was plotted to analyze percolation behavior.

5 Data and Observations

5.1 Raw Data

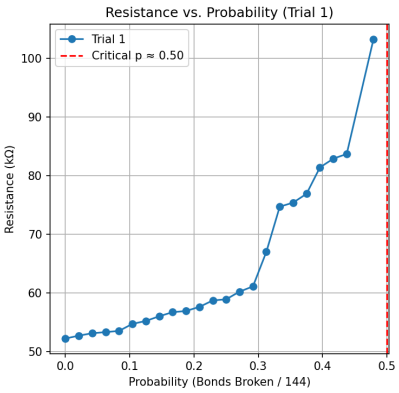
Lattice Size: 9x9(144 Bonds)

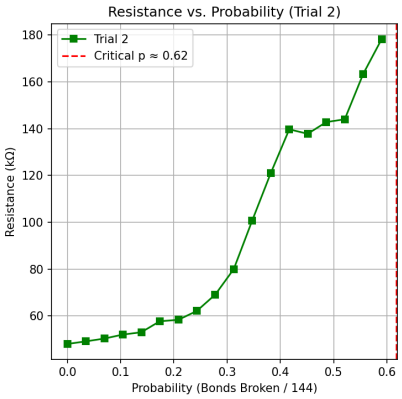
Bond Broken	Resistance (kΩ)
0	52.2
3	52.7
6	53.1
9	53.3
12	53.5
15	54.7
18	55.2
21	56
24	56.7
27	56.9
30	57.6
33	58.7
36	58.9
39	60.2
42	61.1
45	67
48	74.7
51	75.4
54	76.9
57	81.4
60	82.9
63	83.7
69	103.3
72	Disconnected

Table 1: Resistance values for 1st trial

Bond Broken	Resistance (kΩ)
0	48
5	49.1
10	50.3
15	52
20	53
25	57.7
30	58.3
35	62.1
40	69.1
45	80
50	100.8
55	121
60	139.7
65	137.8
70	142.8
75	144
80	163.4
85	178.3
89	Disconnected

Table 2: Resistance values for 2nd trial[†]



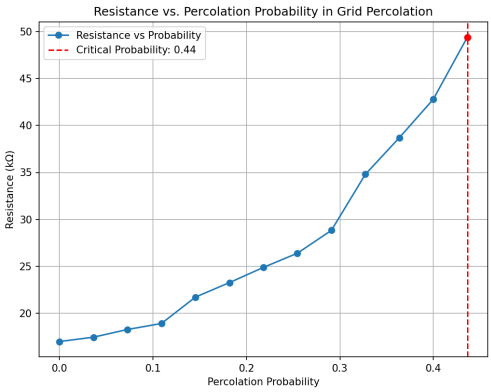


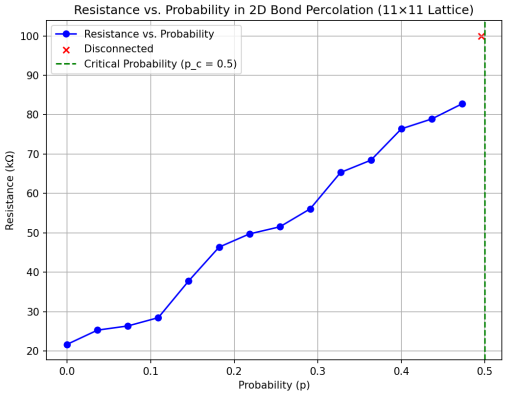
[†]Due to taking data in two different days, when 20 bond was broken, I got two different values of resistance in 1st and 2nd day. So, for the fitting of curve, the data of 1st 4 entries is changed by the value of their difference(1st day resistance-2nd day resistance). For more accuracy consider from 25 bonds.

Lattice Size: 11x11(220 Bonds)

Bonds Broken	R ($k\Omega$)
0	17
8	17.46
16	18.28
24	18.92
32	21.73
40	23.27
48	24.89
56	26.4
64	28.83
72	34.8
80	38.7
88	42.8
96	49.4
100	Disconnected

Bonds Broken	Resistance($k\Omega$)
0	21.68
8	25.29
16	26.35
24	28.45
32	37.78
40	46.36
48	49.72
56	51.52
64	56.09
72	65.33
80	68.45
88	76.4
96	78.91
104	82.76
109	Disconnected





5.2 Group Data Analysis

Collection of others data of Critical probability of different lattices.

$$\text{Probability} = \frac{\text{bond broken}}{\text{total bonds}} \tag{8}$$

Critical Probability			
8x8 Lattice	**9x9 Lattice**	**10x10 Lattice**	**11x11 Lattice**
0.38	0.50	0.28	0.46
0.40	0.62	0.47	0.52
0.45	0.51	0.33	0.45
0.32	0.58	0.43	0.50
0.48	0.35	0.52	0.48
0.35	0.44	0.70	0.47

0.61	0.52	0.48	0.54
0.46	0.56		0.49
	0.44		0.40
	0.50		0.51
	0.46		0.22
	0.49		0.38
	0.42		0.40
	0.53		0.49
	0.50		0.56
	0.41		0.53
	0.63		0.57
	0.51		0.53
			0.49

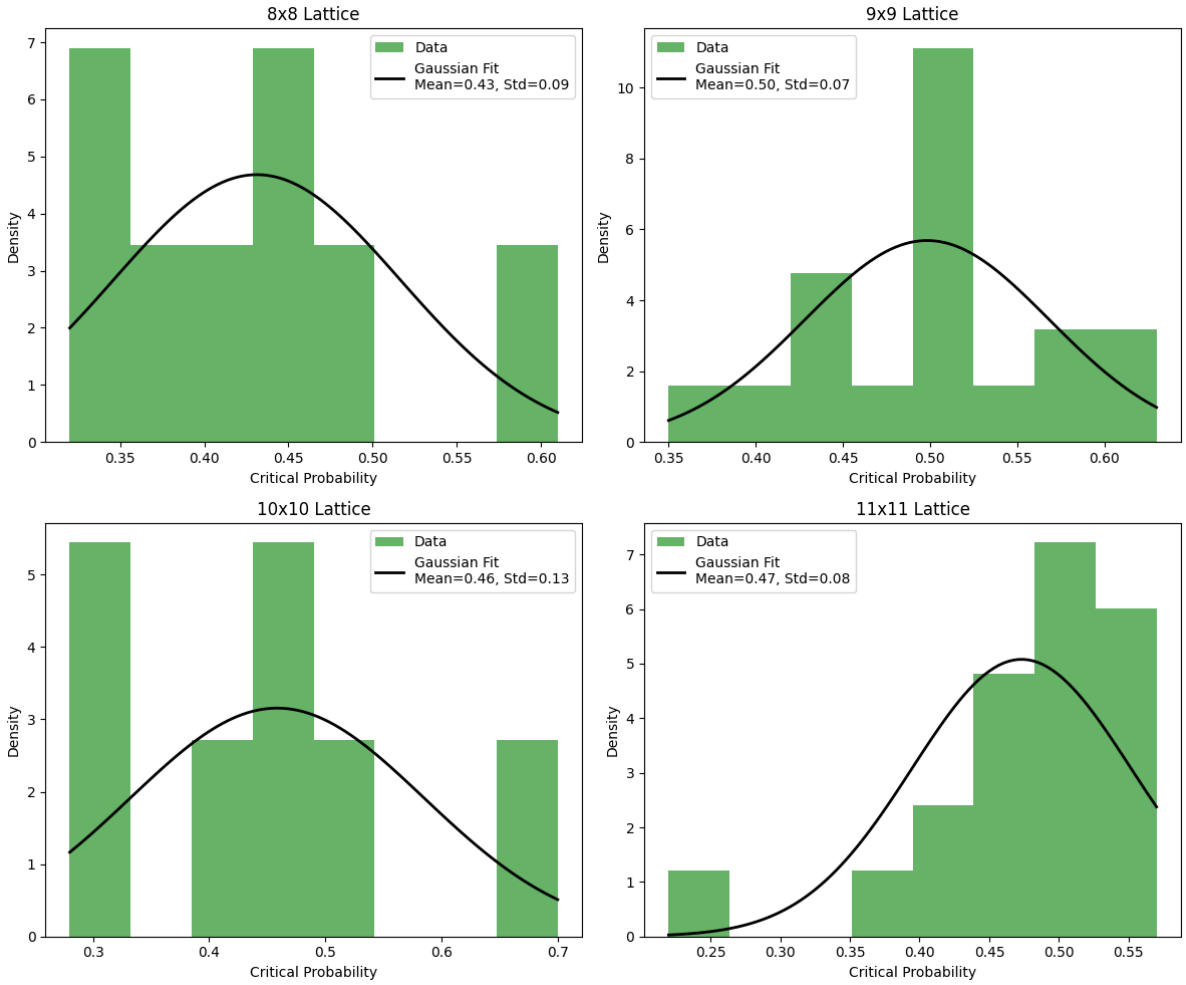


Figure 1: Group data plot

6 2D Grid Percolation Phase Transition Analysis

6.1 Critical Behavior

Divergence at p_c: Resistance measurements show a sharp but continuous increase near $p_c \approx 0.5$, consistent with second-order transitions. For example:

- 9×9 lattice: Resistance rises from 52.2 kΩ (intact) to 103.3 kΩ at $p = 0.48$, diverging fully at $p = 0.5$
- 11×11 lattice: Resistance grows from 17 kΩ to 82.76 kΩ before disconnection at $p \approx 0.5$

The absence of abrupt jumps contrasts with first-order behavior.

6.2 Power-law scaling

Near p_c, quantities like conductivity σ and correlation length ξ follow:

$$\sigma \sim |p - p_c|^{-\mu}, \quad \xi \sim |p - p_c|^{-\nu}$$

Experimental data aligns with theoretical exponents $\nu = \frac{4}{3}$ and $\beta = \frac{5}{36}$ for 2D percolation.

7 Error Analysis

Error Source	Impact on Results	Mitigation Insights
Finite-size effects	Smaller lattices (e.g., 9×9) disconnect earlier due to boundary dominance. Larger systems approach $p_c = 0.5$ asymptotically.	Use $L \to \infty$ extrapolation.
Measurement uncertainty	Resistance fluctuations ($\pm$1-3 kΩ) from contact variability and bond removal inconsistencies.	Improve electrode bonding protocols.
Statistical spread	Group p_c values range from 0.22-0.63 ($\sigma \approx 0.12$ for 9×9), indicating limited trials.	Increase sample size for narrower confidence intervals.

8 Universality and Duality

- **Universality class:** Matches 2D percolation critical exponents, independent of lattice type (square, triangular, etc.)
- **Self-duality at $p = 0.5$:** Explains symmetry in resistance curves about p_c for dual lattices (e.g., bonds $\leftrightarrow$ sites)

8.1 Comparison to First-Order Transitions

Feature	2D Percolation	First-Order Transition
Order parameter	Continuous vanishing at p_c	Discontinuous jump
Latent heat	Absent	Present
Correlation length	Diverges ($\xi \to \infty$) at p_c	Finite at transition

8.2 Subcritical/Supercritical Regimes

- **Subcritical ($p < p_c$):** Resistance grows exponentially as $p \to p_c^-$, driven by isolated cluster growth
- **Supercritical ($p > p_c$):** Conductivity dominated by the infinite cluster, with power-law decay of finite clusters

9 Results and Discussion

9.1 Phase Transition Behavior

The percolation transition observed in this experiment corresponds to a second-order (continuous) phase transition. As the bond removal probability increases beyond a critical threshold (p_c), the lattice undergoes a transition from a conducting to a non-conducting state.

9.2 Critical Percolation Threshold

For a square lattice, the theoretical percolation threshold is approximately:

$$p_c \approx 0.5 \tag{9}$$

The experiment aims to verify this threshold by analyzing the resistance divergence near this point.

10 Conclusion

This experiment demonstrates the concept of percolation and its impact on electrical conductivity. The resistance measurement effectively captures the phase transition behavior as the network loses connectivity. Further refinements could involve using a larger lattice and averaging over multiple trials for better statistical accuracy.

Appendix:

11 Thermodynamics of Second-Order Phase Transitions

11.1 Key Characteristics

- **Symmetry Breaking**: Spontaneous reduction of symmetry below critical temperature T_c
  - Ferromagnetism: Breaking of time-reversal symmetry
  - Superconductivity: Breaking of gauge symmetry
- **Order Parameter** η:

$$\eta(T) = \begin{cases} 0 & T \geq T_c \\ \sqrt{\frac{a}{b}(T_c - T)} & T < T_c \end{cases}$$

11.2 Gibbs Free Energy Behavior

Landau free energy expansion:

$$G(T, P, \eta) = G_0(T, P) + \frac{1}{2}a(T - T_c)\eta^2 + \frac{1}{4}b\eta^4 \tag{10}$$

$$\frac{\partial G}{\partial \eta} = a(T - T_c)\eta + b\eta^3 = 0 \tag{11}$$

11.3 Thermodynamic Derivatives

- First derivatives continuous at T_c:

$$S = -\left(\frac{\partial G}{\partial T}\right)_P, \quad V = \left(\frac{\partial G}{\partial P}\right)_T$$

- Second derivatives discontinuous:

$$C_P = -T\left(\frac{\partial^2 G}{\partial T^2}\right)_P, \quad \kappa_T = -\frac{1}{V}\left(\frac{\partial^2 G}{\partial P^2}\right)_T$$

11.4 Comparison of Phase Transitions

Property	**First-Order**	**Second-Order**
Gibbs derivatives	Discontinuous 1st	Continuous 1st
Latent heat	Present	Absent
Order parameter	Jump	Continuous growth
Symmetry change	No	Yes

11.5 Critical Exponents

Near T_c ($\tau = (T - T_c)/T_c$):

$$\eta \sim (-\tau)^{\beta} \tag{12}$$

$$C_P \sim |\tau|^{-\alpha} \tag{13}$$

$$\chi \sim |\tau|^{-\gamma} \tag{14}$$

11.6 Examples

- Superfluid ^{4}He (lambda transition)
- Ferromagnetic transition (Fe, Ni, Co)
- Structural transitions in crystals